SYMBIOSIS INSTITUTE OF TECHNOLOGY, PUNE

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Subject: Web Application and Development	**Sem: III AY: 2026-27**
EXPERIMENT No: 9	**Assignment No: 9**
TITLE: To configure a local document-oriented NoSQL database system using MongoDB, mapping system paths, and managing schema administration shell variables.	

Objective

To install MongoDB Community Server and MongoDB Shell (mongosh)/Compass, and to perform basic database creation and deletion operations.

Theory

MongoDB is a document-oriented NoSQL database that stores data as flexible, JSON-like BSON documents grouped into collections (analogous to tables), rather than rigid rows/columns. mongosh is the modern MongoDB Shell for interacting with the database via commands; MongoDB Compass is the official GUI. In MongoDB, a database is not physically created until it contains at least one document — the use <dbname> command simply switches context to (or creates) a database, but it only persists once data is inserted into a collection.

Software / Hardware Requirements

MongoDB Community Server installer, MongoDB Compass, a Windows/Mac/Linux machine with admin rights.

Step-by-Step Procedure

1. Download the MongoDB Community Server installer for your OS from mongodb.com/try/download/community.
2. Run the installer, choose 'Complete' setup type, and ensure 'Install MongoDB as a Service' is checked so it starts automatically.
3. Also install MongoDB Compass (bundled with the installer, or download separately) for a graphical interface.
4. Verify installation by opening a terminal/command prompt and running mongod --version and mongosh --version.
5. Start the MongoDB service if not already running: on Windows check Services app for 'MongoDB Server'; on Mac/Linux run sudo systemctl start mongod (or brew services start mongodb-community).
6. Open a terminal and launch the shell by typing mongosh, which connects to the default local server at mongodb://localhost:27017.
7. List existing databases with the show dbs command to see the default admin, config and local databases.

```
show dbs
```

8. Create/switch to a new database named collegeDB using the use command.

```
use collegeDB
```

9. Run show dbs again — notice collegeDB does NOT yet appear because it is empty (MongoDB creates the database lazily).

10. Insert one document into a collection to actually materialize the database.

```
db.students.insertOne({ name: "Aditi Sharma", rollNo: 101, branch: "CSE" })
```

11. Run show dbs again and confirm collegeDB now appears in the list, since it contains data.

12. Verify the current database context with the db command, and list its collections with show collections.

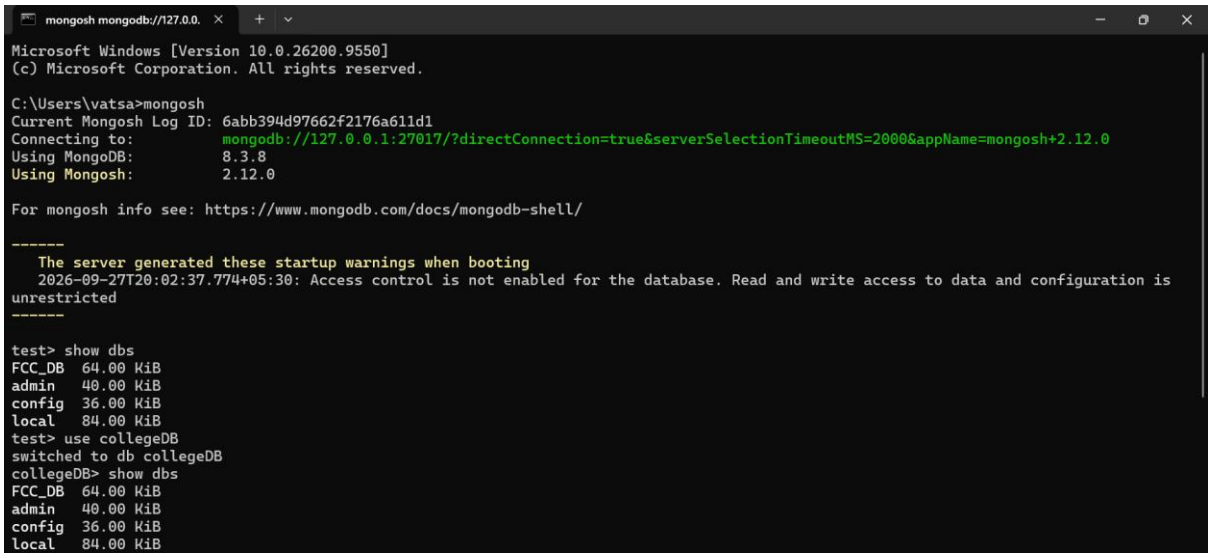
```
db
show collections
```

13. Drop the database entirely using db.dropDatabase() while collegeDB is the active context, then confirm with show dbs that it no longer appears.

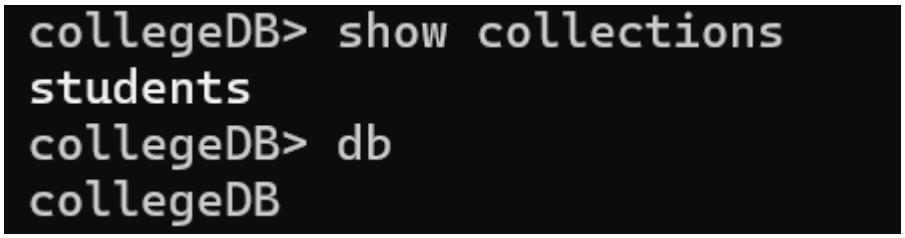
```
use collegeDB
db.dropDatabase()
show dbs
```

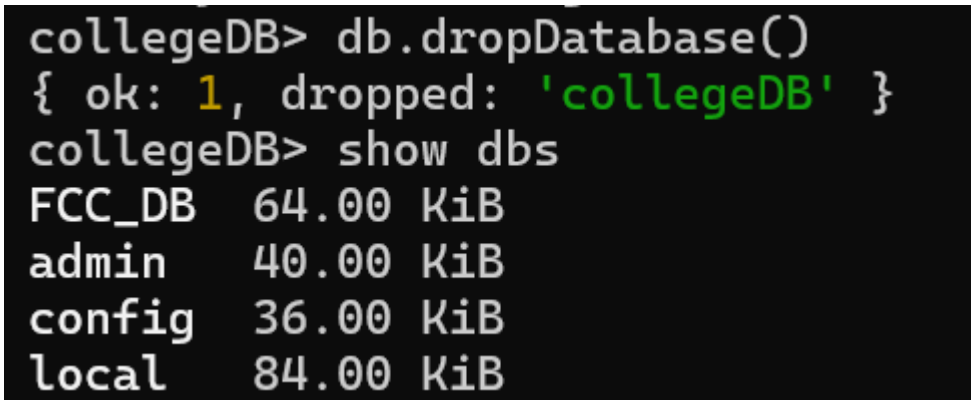
14. Open MongoDB Compass, connect to localhost:27017, and repeat the create/drop database exercise through the GUI to compare it with the shell workflow.

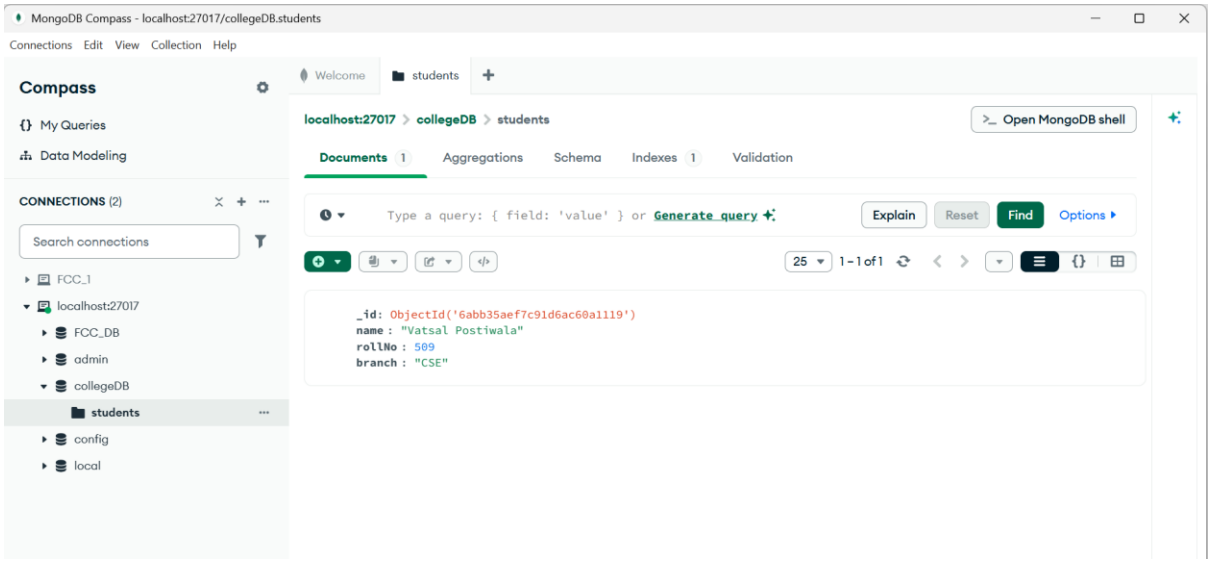
OUTPUT:

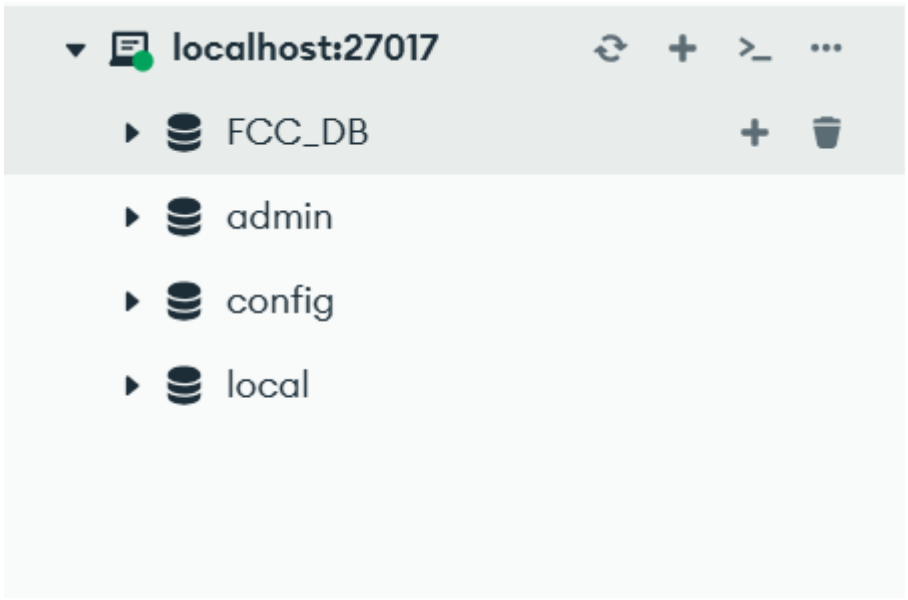

mongosh mongodb://127.0.0.
Microsoft Windows [Version 10.0.26200.9550]
(c) Microsoft Corporation. All rights reserved.
C:\Users\vatsa>mongosh
Current Mongosh Log ID: 6abb394d97662f2176a611d1
Connecting to: mongodb://127.0.0.1:27017/?directConnection=true&serverSelectionTimeoutMS=2000&appName=mongosh+2.12.0
Using MongoDB: 8.3.8
Using Mongosh: 2.12.0
For mongosh info see: https://www.mongodb.com/docs/mongodb-shell/
The server generated these startup warnings when booting
2026-09-27T20:02:37.774+05:30: Access control is not enabled for the database. Read and write access to data and configuration is unrestricted
test> show dbs
FCC_DB 64.00 KiB
admin 40.00 KiB
config 36.00 KiB
local 84.00 KiB
test> use collegeDB
switched to db collegeDB
collegeDB> show dbs
FCC_DB 64.00 KiB
admin 40.00 KiB
config 36.00 KiB
local 84.00 KiB


collegeDB> db.students.insertOne({ name: "Vatsal Postiwala", rollNo: 509, branch: "CSE" })
{
acknowledged: true,
insertedId: ObjectId('6abb399397662f2176a611d2')
}
collegeDB> show dbs
FCC_DB 64.00 KiB
admin 40.00 KiB
collegeDB 8.00 KiB
config 36.00 KiB
local 84.00 KiB


collegeDB> show collections
students
collegeDB> db
collegeDB

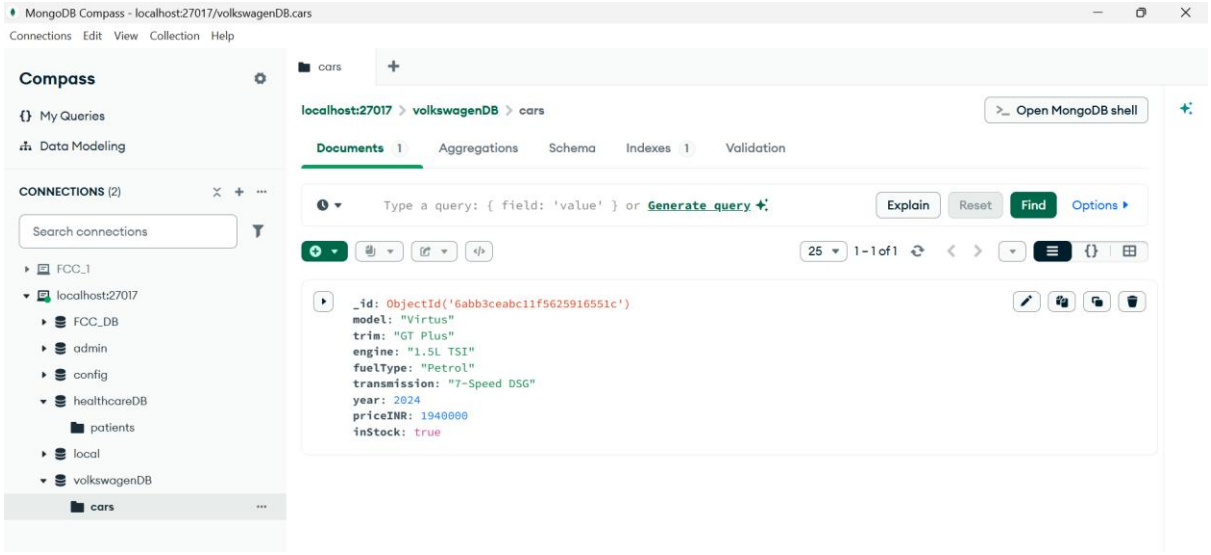

collegeDB> db.dropDatabase()
{ ok: 1, dropped: 'collegeDB' }
collegeDB> show dbs
FCC_DB 64.00 KiB
admin 40.00 KiB
config 36.00 KiB
local 84.00 KiB


MongoDB Compass - localhost:27017/collegeDB.students
Connections Edit View Collection Help
Compass
My Queries
Data Modeling
CONNECTIONS (2)
Search connections
FCC_1
localhost:27017
FCC_DB
admin
collegeDB
students
config
local
Welcome
students
localhost:27017 > collegeDB > students
Open MongoDB shell
Documents 1
Aggregations
Schema
Indexes 1
Validation
Type a query: { field: 'value' } or Generate query
Explain
Reset
Find
Options
25
1 - 1 of 1
_id: ObjectId('6abb35aef7c91d6ac60a1119')
name : "Vatsal Postiwala"
rollNo : 509
branch : "CSE"


localhost:27017
FCC_DB
admin
config
local

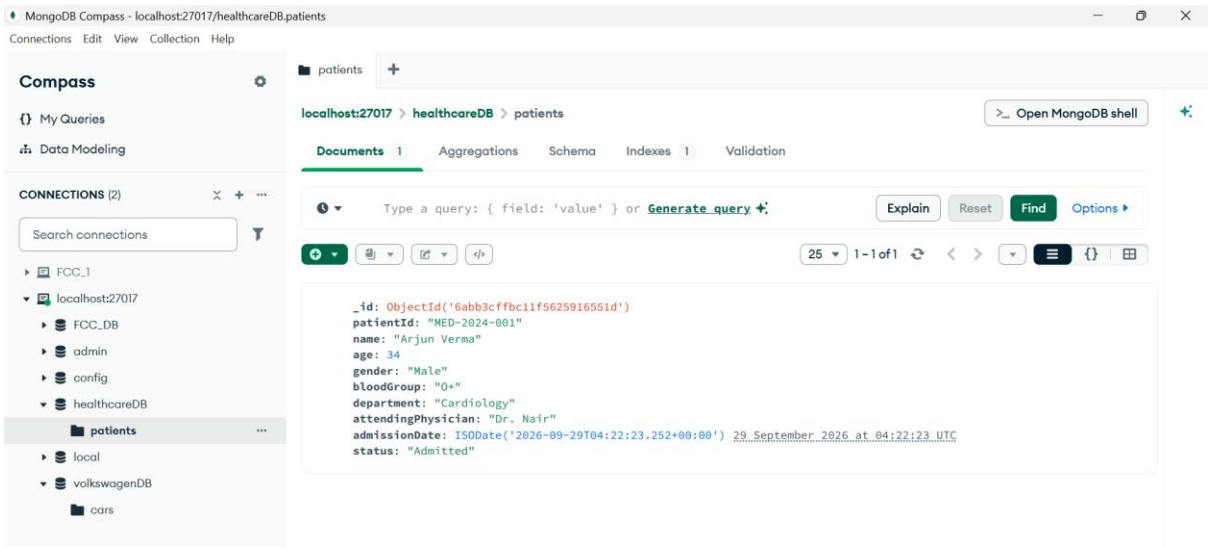
Create volkswagen DB:

```
test> use volkswagenDB
switched to db volkswagenDB
volkswagenDB> db.cars.insertOne({
|    model: "Virtus",
|    trim: "GT Plus",
|    engine: "1.5L TSI",
|    fuelType: "Petrol",
|    transmission: "7-Speed DSG",
|    year: 2024,
|    priceINR: 1940000,
|    inStock: true
| })
|
{
  acknowledged: true,
  insertedId: ObjectId('6abb3ceabc11f5625916551c')
}
```

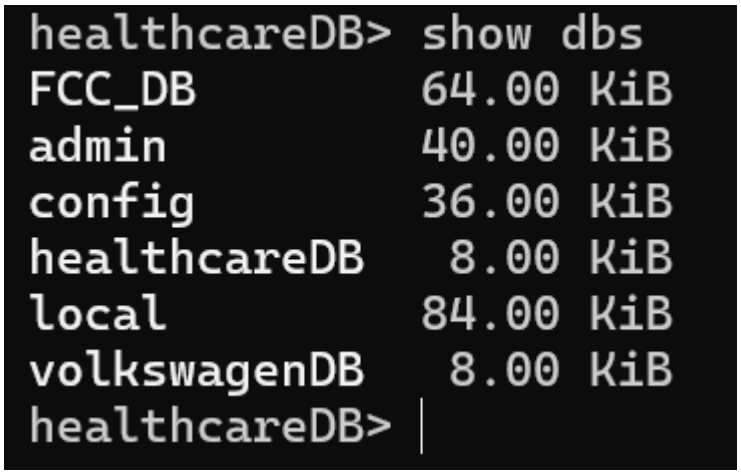


Create HealthCare DB:

```
volkswagenDB> use healthcareDB
switched to db healthcareDB
healthcareDB> db.patients.insertOne({
|   patientId: "MED-2024-001",
|   name: "Arjun Verma",
|   age: 34,
|   gender: "Male",
|   bloodGroup: "O+",
|   department: "Cardiology",
|   attendingPhysician: "Dr. Nair",
|   admissionDate: new Date(),
|   status: "Admitted"
| })
{
  acknowledged: true,
  insertedId: ObjectId('6abb3cffbc11f5625916551d')
}
```



Show dbs :


healthcareDB> show dbs
FCC_DB 64.00 KiB
admin 40.00 KiB
config 36.00 KiB
healthcareDB 8.00 KiB
local 84.00 KiB
volkswagenDB 8.00 KiB
healthcareDB>